

Mark 20.5 /100  
25

Student Name: \_\_\_\_\_

Student ID: \_\_\_\_\_

**Answer All Questions:**

Instructor use only

| Outcomes | 1 | 2  | 3 | 4  | 5  | 6 |
|----------|---|----|---|----|----|---|
| Covered  | ✓ | NA | ✓ | NA | NA | ✓ |

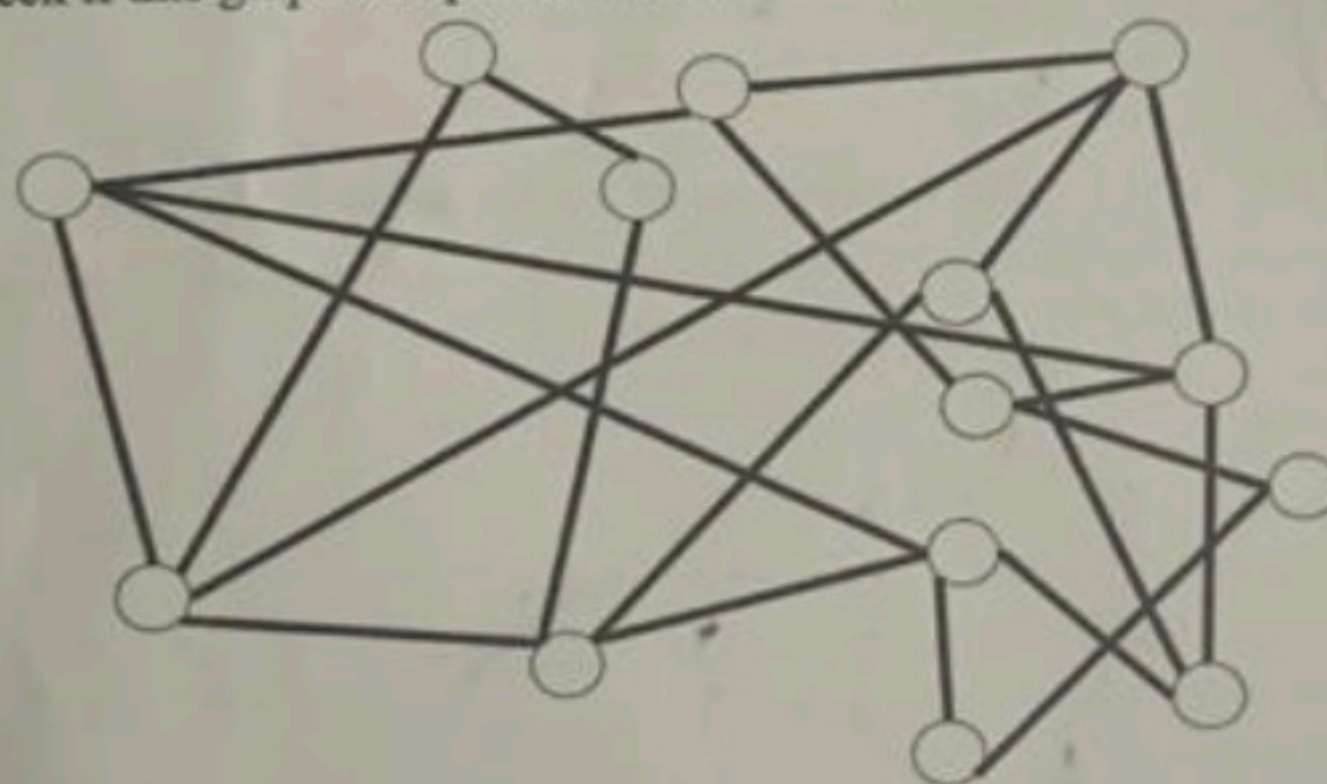
| Question        | 1 | 2 | 3 | 4 | 5 | 6 |
|-----------------|---|---|---|---|---|---|
| Covered Outcome | - | - | 6 | 6 | 1 | 6 |
| Allocated mark  | 2 | 4 | 6 | 8 | 3 | 2 |
| Earned mark     |   |   |   |   |   |   |

| Total |    |      |
|-------|----|------|
| 1     | 6  | Exam |
| 3     | 16 | 19   |

**Instructions:** Answer all the following questions and avoid overwriting using of a pencil.

**Question 1:** Check if this graph is bipartite or not, and why?:

[ 2 Marks]

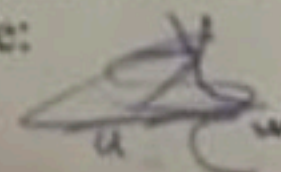


The answer: not bipartite The reason is: the are vertices in the same level

**Question 2:** [ 2.5+1.5 Marks]

a) Fill in the blanks

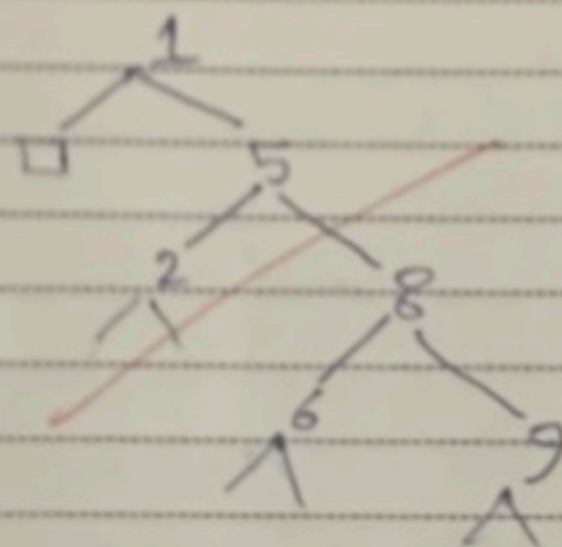
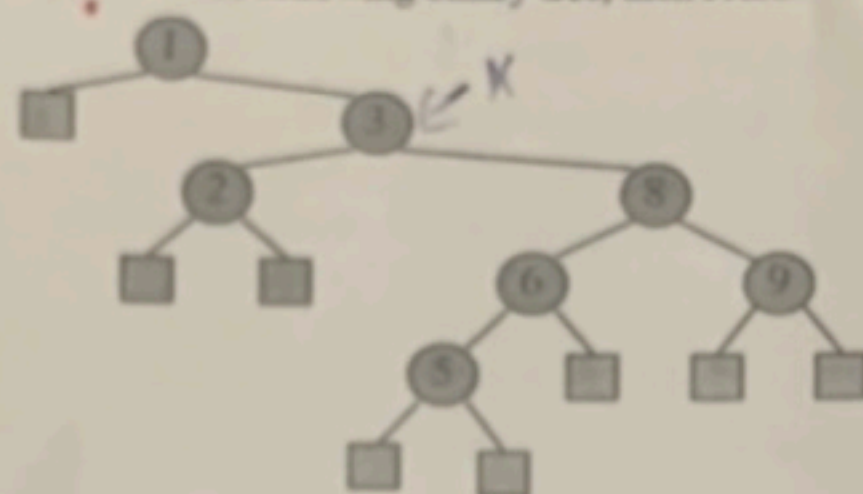
- InsertItem to dictionary takes  $O(1)$  time.
- Bottom-up heap construction runs in  $O(n)$  time.
- In the binary search tree: Let u, v, and w be three nodes such that u is in the left subtree of v and w is in the right subtree of v. Therefore we have:



Question 4: answer the following

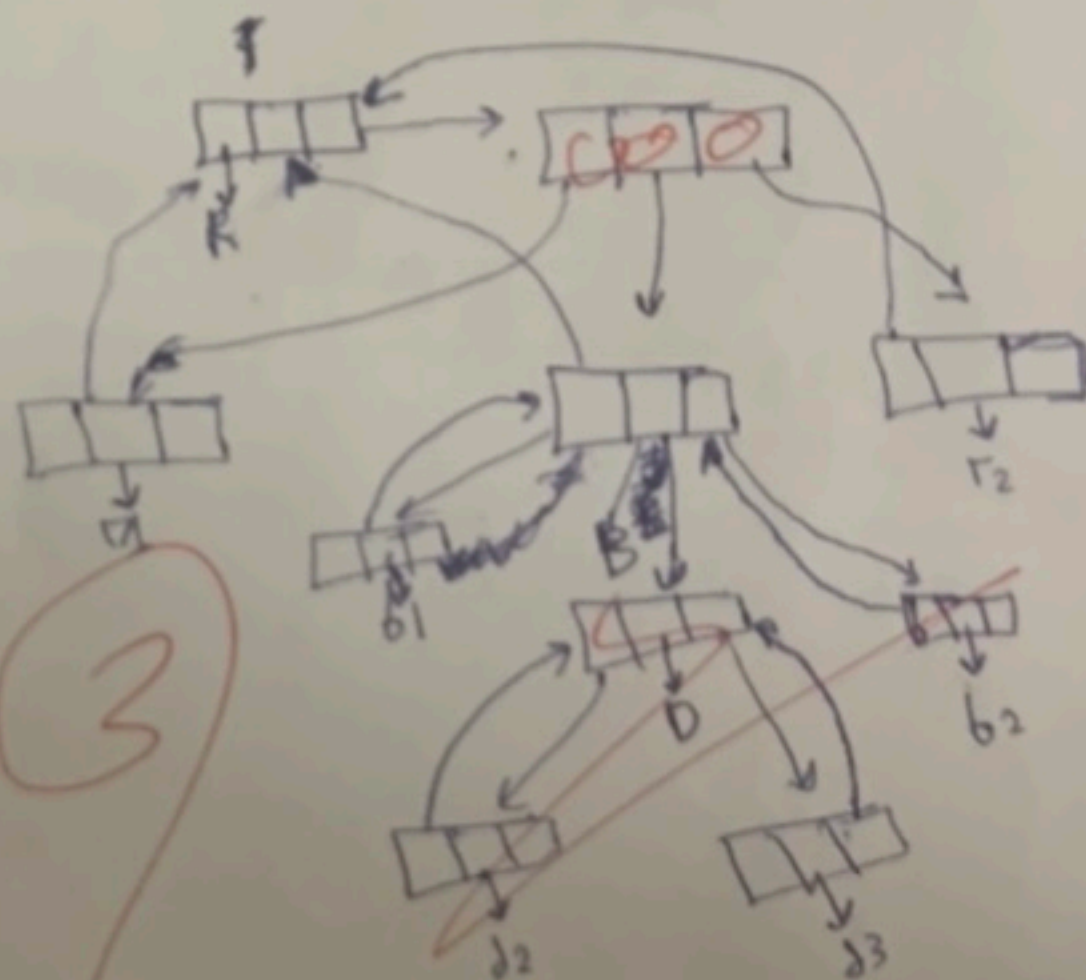
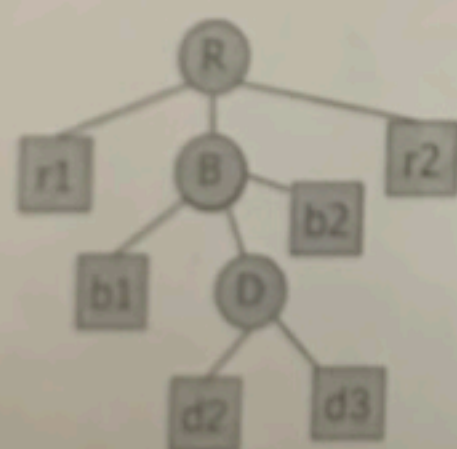
[2+4+2 pts]

a) Remove the node 3 from the following binary tree, then redraw the tree after deletion.

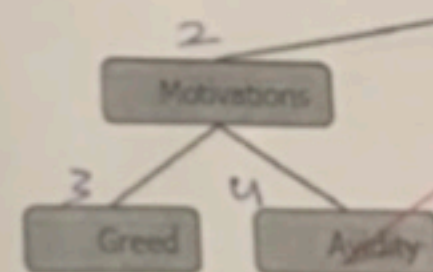


b) Design the Linked Data Structure that represents the following Tree structure.

The answer



c) Write the sequence of tree the figure:



Question 5: Discuss all available

- ① Changing
- ② Learning
- ③ Double

Example on changing

| id | name | link   |
|----|------|--------|
| 1  | 6    |        |
| 2  | 7    | Linked |
| 3  | 8    |        |

Question 6: Count the Total

Algorithm arrayMax(A, n)  
 currentMax ← A[0]  
 Temp ← A[0]  
 for i ← 1 to n - 1  
 if A[i] > currentMax  
 currentMax ← A[i]  
 { increment count }  
 return currentMax

7n + 1

Good luck: Prof. Dr. M

$$\text{key}(u) \leq \text{key}(w) \leq \text{key}(v)$$

- The space used by a Queue with a Singly Linked list is  $O(n)$
  - FindElement in binary tree takes  $O(n)$
- b) Put T or F for each of the following statements.
- ( F ) In order to compare two algorithms, the same hardware and software environments must be used without any consideration of inputs type and its size.
  - ( T ) Theoretical analysis allows us to evaluate the speed of an algorithm independent of the hardware/software environment.
  - ( T ) The total of primitive operations represents the Worst case of an algorithm.

Question 3:

① [4+2 pts] ②

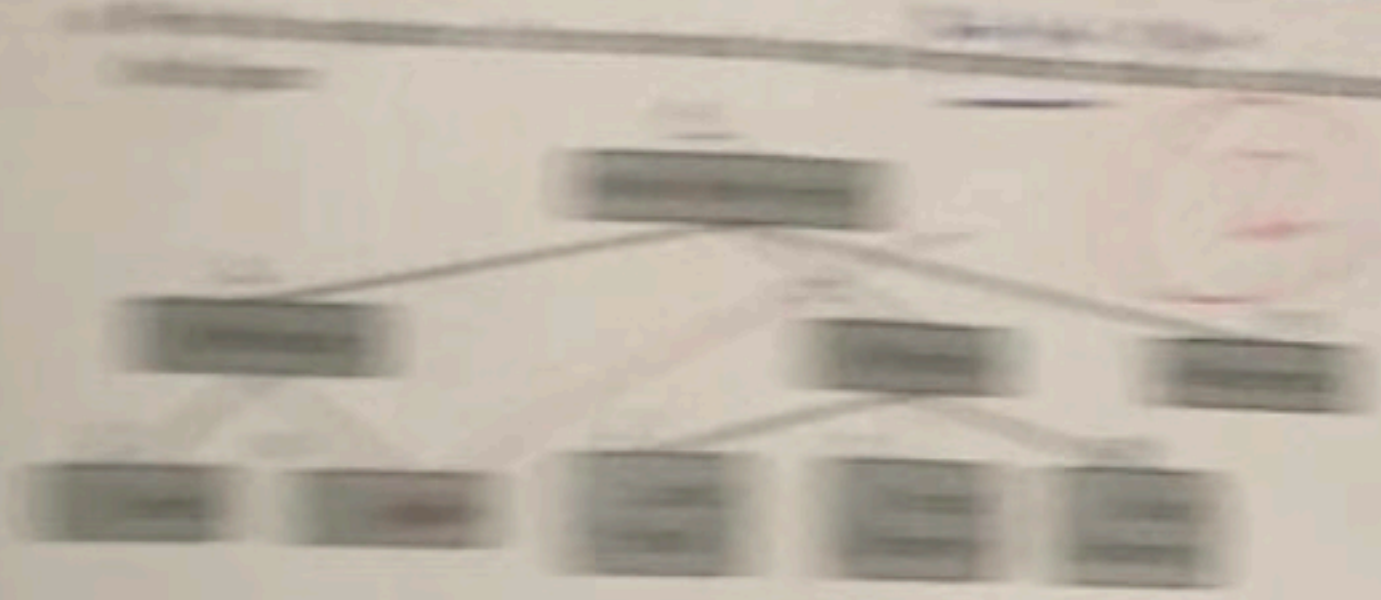
- a) what is the objective of Upheap, Downheap algorithms? when to use? then what time required to run UpHeap and DownHeap?

The objective of Upheap is for example when you insert a new element then may be the heap order become wrong so we use up heap to fix the order. On the other hand we need down heap when we delete an element so we must reorder the tree using down heap.

The time is  $O(n)$

- b) When you can say that the hash function is "good"?

when you address your items without collision



السؤال الاول كان ترافيرس ان  
اوردر اظن  
الثاني: يتكلم عن الهاش فنكشن  
الثالث: حساب اوبريشن